



LIGHT -02

Class 10 - Science

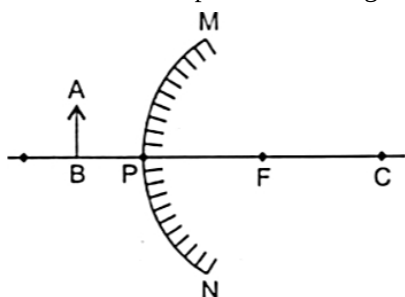
1. Draw ray diagram showing the image formation by a concave mirror when an object placed between pole and focus of the mirror. [2]
2. An object is placed at a distance of 12 cm in front of a concave mirror. It forms a real image four times larger than the object. Calculate the distance of the image from the mirror [2]
3. A convex mirror used on a bus has a focal length of 200 cm. If a scooter is located at 100 cm. from this mirror find the position, nature and magnification of the image formed in the mirror. [2]
4.
 - i. A security mirror used in a big showroom has radius of curvature 5 m. If a customer is standing at a distance of 20 m from the cash counter, find the position, nature and size of the image formed in the security mirror. [2]
 - ii. Neha visited a dentist in his clinic. She observed that the dentist was holding an instrument fitted with a mirror. State the nature of this mirror and reason for its use in the instrument used by dentist.
5. A concave mirror produces three times magnified real image of an object placed at 10 cm in front of it. Where is the image located ? [2]
6. Define: [2]
 - i. Focus of a concave mirror
 - ii. Focus of a convex mirror
7. State the two laws of reflection of light. [2]
8. Name the type of mirror used in a solar furnace. How can high temperature be achieved by this device? [2]
9. Size of image of an object by a mirror having a focal length of 20cm is observed to be reduced to 1/3rd of its size. At what distance the object has been placed from the mirror? What is the nature of the image and the mirror? [2]
10. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image. [2]
11. A 4 cm tall object is placed on the principal axis of convex lens. The distance of the object from the optical centre of the lens is 12 cm and its sharp image is formed at a distance of 24 cm from it on a screen on the other side of the lens. If the object is now moved a little away from the lens, in which way (towards the lens or away from the lens) will he have to move the screen to get a sharp image of the object on it again? How will the magnification of the image be affected? [2]
12. A concave mirror is used for image formation for different positions of an object. What inferences can be drawn about the following when an object is placed at a distance of 10 cm from the pole of a concave mirror of focal length 15 cm? [2]
 - i. Position of the image
 - ii. Size of the image

iii. Nature of the image

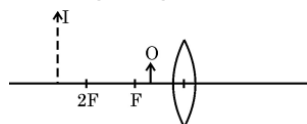
Draw a labelled ray diagram to justify your inferences.

13. We wish to obtain an erect image of an object, using a concave mirror of focal length 15 cm. What should be the range of distance of the object from mirror? What is the nature of image? Is the image larger or smaller than the object? Draw a ray diagram to show the image formation in this case. [2]
14. Draw ray diagram showing the image formation by a concave mirror when an object placed between focus and centre of curvature of the mirror. [2]
15. What is mirror formula ? Does this formula hold good for a plane mirror ? [2]
16. A student wants to project the image of a candle flame on a screen 48 cm in front of a mirror by keeping the flame at a distance of 12 cm from its pole. [2]
 - i. Suggest the type of mirror he should use.
 - ii. Find the linear magnification of the image produced.
 - iii. How far is the image from its object?
 - iv. Draw a ray diagram to show the image formation in this case.
17. The image formed by a spherical mirror is real, inverted and is of magnification -2. If the image is at a distance of 30 cm from the mirror, where is the object placed? Find the focal length of the mirror. List two characteristics of the image formed if the object is moved 10 cm towards the mirror. [2]
18. Write the spherical mirror formula and explain the meaning of each symbol used in it. [2]
19. Draw ray diagram to show the principal focus of a: [2]
 - a. Concave Mirror
 - b. Convex Mirror

20. a. The figure show a convex mirror. C is its centre of curvature and F is its focus. Draw two rays from A and hence locate the position of image of the object AB. Label the image as **AB**. [2]



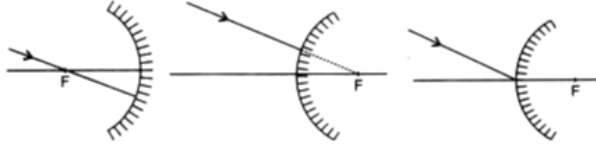
- b. State three characteristics of the image formed.
- c. State one application of a convex mirror.
21. i. An object 1 cm high produces a real image 1.5 cm high when placed at a distance of 15cm from a concave mirror. Calculate the position of the image and the magnification. [2]
 - ii. Write two uses of concave mirrors.
22. A truck uses a convex mirror as view finder whose radius of curvature is 2.0 m. A maruti car is coming behind the truck at a distance of 10 m. What will be the position of the image of the car and size of the image of the car when observed by the driver of the truck through the convex mirror? [2]
23. Find the focal length of a convex mirror whose radius of curvature is 32 cm. [2]
24. The diagram given below shows an object O and its image I. [2]



Without actually drawing the ray diagram, state the following:

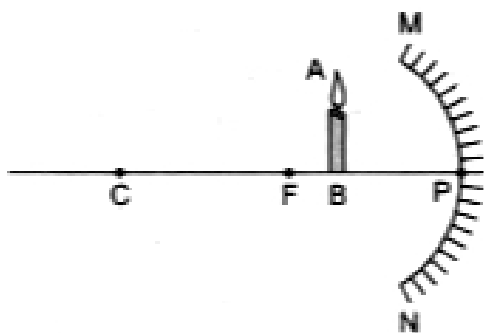
- i. Type of lens (Converging/Diverging)
- ii. Name two optical instruments where such an image is obtained.
- iii. List three characteristics of the image formed if this lens is replaced by a concave mirror of focal length f and an object is placed at a distance $\frac{f}{2}$ in front of the mirror.

25. Draw the following diagram, in which a ray of light is incident on a concave/convex mirror, on your answer sheet. Show the path of this ray, after reflection, in each case. [2]

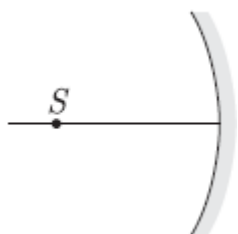


26. Why do we prefer a convex mirror as a rear-view mirror in vehicles? [2]
27. How do you find the rough focal length of a convex lens? Is the same method applicable to a concave lens? [2]
28. State the mirror formula for determining the focal length of spherical mirrors write the meanings of the symbols used. An object is placed at a distance of 25 cm. from a concave mirror of focal length 15 cm. Calculate the distance of the image from the mirror. [2]
29. A concave mirror produces three times magnified (enlarged) real image of an object placed at 10 cm in front of it. Where is the image located? [2]
30. A concave mirror produces three times magnified (enlarged) real image of an object 10 cm in front of it. Where is the image located? [2]
31. A spherical mirror produces an image of magnification -1 on a screen placed at a distance of 50 cm from the mirror. [2]
- i. Write the type of mirror.
 - ii. Find the distance of the image from the object.
 - iii. What is the focal length of the mirror?
 - iv. Draw the ray diagram to show the image formation in this case.
32. Name the type of mirror used in the following situations: [2]
- (a) Headlights of a car
 - (b) Side/rear-view mirror of a vehicle.
 - (c) Solar furnace.
- Support your answer with reason.
33. Draw ray diagram showing the image formation by a concave mirror when an object placed at centre of curvature of the mirror. [2]
34. If an object is held at a distance of 60 cm from a convex mirror of focal length 20 cm. At what distance from the convex mirror, should a plane mirror be held, so that images in the two mirrors coincide? [2]
35. The image formed by a concave mirror is observed to be virtual, erect and larger than the object. where should be the position of object? Justify your answer. [2]
36. An object 5.0 cm in length is placed at a distance of 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position of the image, its nature and size. [2]
37. To construct a ray diagram we use two rays which are so chosen that it is easy to know their directions after reflection from the mirror. List two such rays and state the path of these rays after reflection in case of concave mirrors. Use these two rays and draw ray diagram to locate the image of an object placed between pole and focus of a concave mirror. [2]

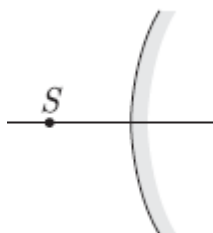
38. a. The figure shows a concave mirror with its pole at P, focus F and centre of curvature C. Draw ray diagram to show the formation of image of object AB. [2]
 b. State three characteristics of the image formed.
 c. Mention one application of image formed in this case.



39. The magnification produced when an object is placed at a distance of 20 cm from a spherical mirror is $+\frac{1}{2}$. [2]
 Where should the object be placed to reduce the magnification to $+\frac{1}{3}$.
 40. The formula for linear magnification of a spherical mirror is $m = \frac{h_2}{h_1} = \frac{-v}{u}$ What u determines the sign of m? [2]
 What is the significance of this sign?
 41. A convex mirror used for rear-view on an automobile has a radius of curvature of 3.00 m. If a bus is located at 5.00 m from this mirror, find the position, nature and size of the image. [2]
 42. On the way from Kanpur to Delhi there were four friends. Sunil was driving the car and saw from his side mirror that the car which was behind their car had met an accident. He suddenly applied the brake even after his friends asked him to leave the situation as it is. But Sunil did not agree and get down of car and persuaded his friends to help the injured. All of them took the injured person to the nearest hospital. After taking first aid from hospital the victim thanked and pleased them for saving his life. Read the above passage and answer the following questions: [2]
 i. Name the type of mirror from which Sunil saw the accident.
 ii. Why this mirror is used as a side mirror in vehicles?
 iii. What can you learn from the Sunil's character?
 43. Name a mirror which can give an erect and enlarged image of an object. [2]
 44. An optical component and an object S is placed along its optic axis. The distance between the object and the component can be vary. [2]



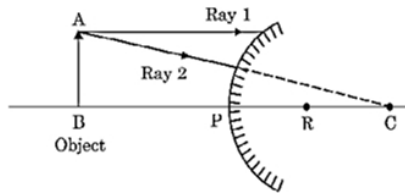
- i. What types of image can be formed in the above diagram?



ii. What type of images can be formed in the above diagram?

iii. Draw the ray diagram for the image formation by a concave mirror when the object is placed at the focus.

45. a. Complete the following ray diagram to show the formation of image: [2]



b. Mention the nature, position and size of the image formed in this case.

c. State the sign of the image distance in this case using the Cartesian sign convention.

46. A student wants to project the image of a candle flame on a screen 60 cm in front of a mirror by keeping the candle at a distance of 15 cm from its pole. [2]

a. Name the type of mirror used.

b. Also calculate:

i. Magnification of the image produced

ii. Distance between object and its image

c. Draw a ray diagram to show the image formation.

47. An object 4 cm in height, is placed at 15 cm in front of a concave mirror of focal length 10 cm. At what distance from the mirror should a screen be placed to obtain a sharp image of the object. Calculate the height of the image. [2]

48. Rohit wants to have an erect image of an object using a converging mirror of focal length 40 cm. [2]

i. Specify the range of distance where the object can be placed in front of the mirror. Give reason for your answer.

ii. Will the image be bigger or smaller than the object?

iii. Draw a ray diagram to show the image formation in this case.

49. "The magnification produced by a spherical mirror is -3." List four information you obtain from this statement about the mirror/image. [2]

50. Draw ray diagram showing the image formation by a concave mirror when an object placed a little beyond centre of curvature of the mirror. [2]

51. Draw ray diagrams showing the image formation by a convex mirror when an object is placed at finite distance from the mirror. [2]

52. An object is kept at a distance of 15cm from a [2]

i. convex mirror

ii. concave lens

iii. Plane mirror.

The focal length of the convex mirror and the concave lens are 10 cm each.

Draw the appropriate ray diagrams, showing the formation of image, in each of the three cases.

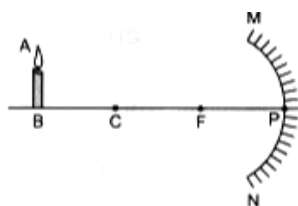
53. Draw ray diagrams showing the image formation by a convex mirror when an object is placed at infinity. [2]

54. A concave mirror of focal length 10 cm is placed at a distance of 35 cm from a wall. How far from the wall an object be placed so that its image formed by mirror falls on the wall? [2]

55. An object of size 7.0 cm is placed 27 cm in front of concave mirror of focal length 18 cm. At what distance should the screen be placed so that a sharp focused image can be obtained ? Find the size and nature of the [2]

image.

56. Two friends Kapil and Rohit were studying in the same class. One day Rohit observed that Kapil was having pain in gums during lunch time. Rohit told Kapil that his father was dentist and asked him to visit his father's clinic. Rohit's father examined Kapil with the help of a mirror and advised him not to eat too many chocolates and soft drinks. Kapil followed the advice of the doctor and soon he got recovered. After that he starts taking care of his mouth, as he washes his mouth properly after every meal and also starts taking a calcium rich diet. Read the given passage and answer the following questions: [2]
- i. Identify the mirror used by the dentist.
 - ii. Name the phenomenon of light by which doctor is able to examine Kapil.
 - iii. What values are shown by doctor, his son and Kapil?
57. A mirror has magnification 0.4, what type of the mirror is that? and what type of the image is formed? [2]
58. What is the nature of the mirror having focal length – 15 cm? [2]
59. Explain why a ray of light passing through the centre of curvature of a concave mirror gets reflected along the same path after reflection. [2]
60. Name the type of mirror used in the side/rear-view mirror of a vehicle. Support your answer with reason. [2]
61. Draw ray diagram in each of the following cases to show what happens after reflection to the incident ray when [2]
- i. it is parallel to the principal axis and falling on a convex mirror.
 - ii. it is falling on a concave mirror while passing through its principal focus.
 - iii. it is coming oblique to the principal axis and falling on the pole of a convex mirror.
62. Draw ray diagram showing the image formation by a concave mirror when an object placed at infinity. [2]
63. A 6 cm tall object is placed perpendicular to the principal axis of a concave mirror of focal length 30 cm. The distance of the object from the mirror is 45 cm. Use mirror formula to determine the position, nature and size of the image formed. Also draw labelled ray diagram to show the image formation in this case. [2]
64. Name the type of mirror used in the Solar furnace. Support your answer with reason. [2]
65. A student has focussed the image of an object of height 3 cm on a white screen using a concave mirror of focal length 12 cm. If the distance of the object from the mirror is 18 cm, find the values of the following: [2]
- i. Distance of the image from the mirror
 - ii. Height of the image
66. Why do we prefer a convex mirror as back view mirror in vehicles? [2]
67. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image. [2]
68. An object of size 7.0 cm is placed at 27 cm in front of a concave mirror of focal length 18 cm. At what distance from the mirror should a screen be placed, so that a sharp focussed image can be obtained? Find the size and the nature of the image. [2]
69. a. Figure shows a concave mirror with its pole at P, focus F and centre of curvature C. Draw a ray diagram to show the formation of image of an object AB. [2]
b. State three characteristics of the image formed.



70. Why do we prefer a convex mirror as a rear view mirror in vehicles? [2]
71. The image of an object formed by a mirror is real, inverted and is of magnification -1. If the image is at a distance of 40 cm from the mirror, where is the object placed? Where would the image be if the object is moved 20 cm towards the mirror? State reason and also draw ray diagram for the new position of the object to justify your answer. [2]
72. A ray of light is incident on a convex mirror as shown. Redraw the diagram and complete the path of this ray after reflection from the mirror. Mark angle of incidence and angle of reflection on it. [2]

